

2.

```
import java.util.Scanner;
```

```
class StudentDetails {
```

```
    String studentId;
```

```
    String studentName;
```

```
    String department;
```

```
    void displayStudentDetails() {
```

```
        System.out.println("Student ID: " + studentId);
```

```
        System.out.println("Student Name: " + studentName);
```

```
        System.out.println("Department: " + department);
```

```
    }
```

```
}
```

```
class UndergraduateStudent extends StudentDetails {
```

```
    int semester;
```

```
    double cgpa;
```

```
    void displayUGDetails() {
```

```
        System.out.println("Semester: " + semester);
```

```
        System.out.println("CGPA: " + cgpa);
```

```
    }
```

```
}
```

```
class PostgraduateStudent extends StudentDetails {
```

```
    String specialization;
```

```
    String researchTopic;
```

```
void displayPGDetails() {  
    System.out.println("Specialization: " + specialization);  
    System.out.println("Research Topic: " + researchTopic);  
}  
}
```

```
public class Main {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        UndergraduateStudent ug = new UndergraduateStudent();  
  
        System.out.println("Enter Undergraduate Student Details");  
  
        System.out.print("Student ID: ");  
        ug.studentId = sc.nextLine();  
  
        System.out.print("Student Name: ");  
        ug.studentName = sc.nextLine();  
  
        System.out.print("Department: ");  
        ug.department = sc.nextLine();  
  
        System.out.print("Semester: ");  
        ug.semester = sc.nextInt();  
  
        System.out.print("CGPA: ");  
        ug.cgpa = sc.nextDouble();  
    }  
}
```

```
System.out.println("\nUNDERGRADUATE STUDENT");

ug.displayStudentDetails();
ug.displayUGDetails();


sc.nextLine();


PostgraduateStudent pg = new PostgraduateStudent();


System.out.println("\nEnter Postgraduate Student Details");


System.out.print("Student ID: ");
pg.studentId = sc.nextLine();


System.out.print("Student Name: ");
pg.studentName = sc.nextLine();


System.out.print("Department: ");
pg.department = sc.nextLine();


System.out.print("Specialization: ");
pg.specialization = sc.nextLine();


System.out.print("Research Topic: ");
pg.researchTopic = sc.nextLine();


System.out.println("\nPOSTGRADUATE STUDENT");
pg.displayStudentDetails();
pg.displayPGDetails();
```

```
        sc.close();  
    }  
}
```

```
Enter Undergraduate Student Details  
Student ID: 101  
Student Name: ravi  
Department: cse  
Semester: 5  
CGPA: 8.75
```

```
UNDERGRADUATE STUDENT  
Student ID: 101  
Student Name: ravi  
Department: cse  
Semester: 5  
CGPA: 8.75
```

```
Enter Postgraduate Student Details  
Student ID: 201  
Student Name: anitha  
Department: ece  
Specialization: ai  
Research Topic: computer vision
```

```
POSTGRADUATE STUDENT  
Student ID: 201  
Student Name: anitha  
Department: ece  
Specialization: ai  
Research Topic: computer vision
```

```
...Program finished with exit code 0  
Press ENTER to exit console.
```